Imperial College London

Lecture 6

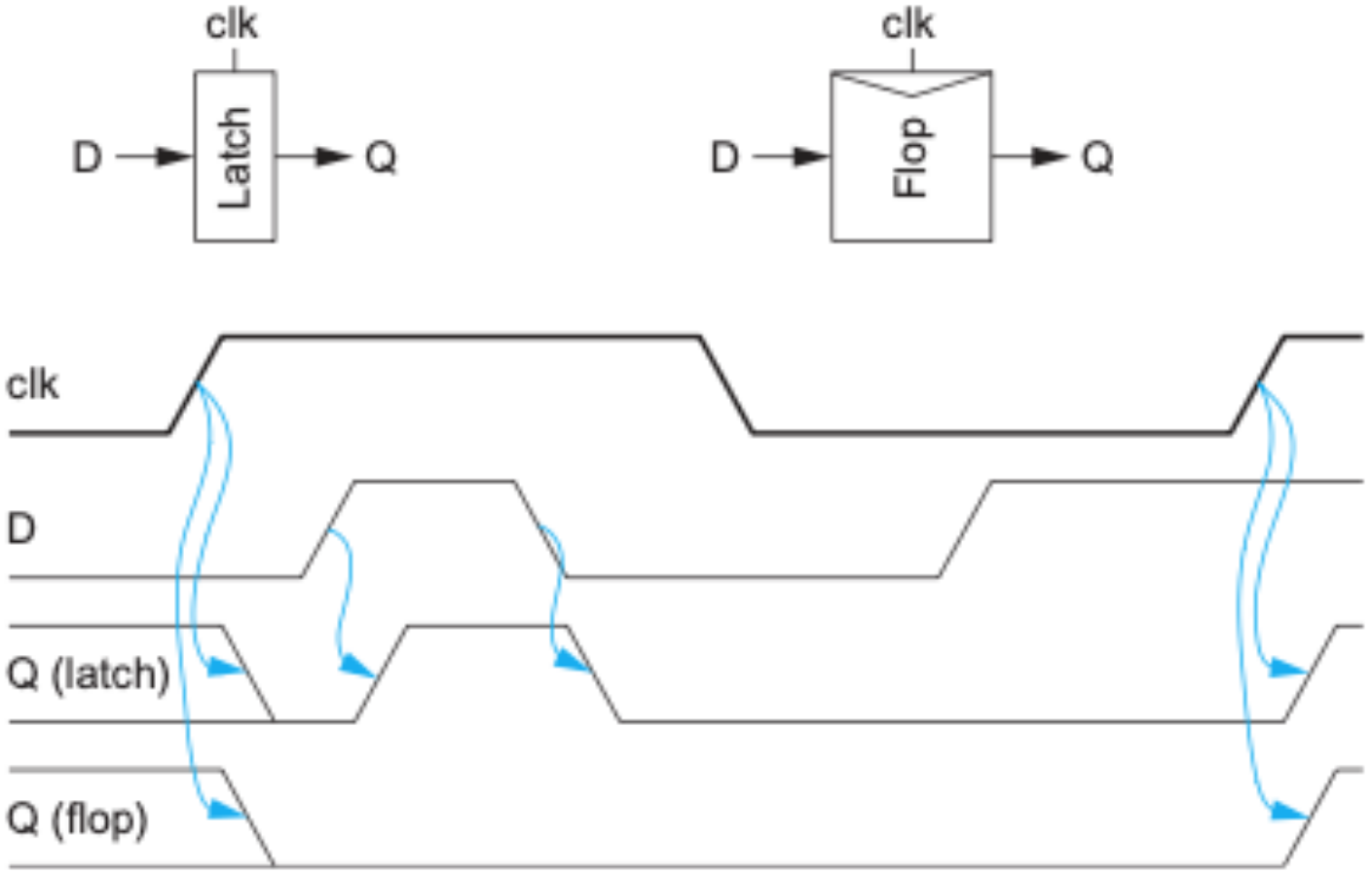
Sequential Circuits

Prof Peter YK Cheung

Department of Electrical & Electronic Engineering

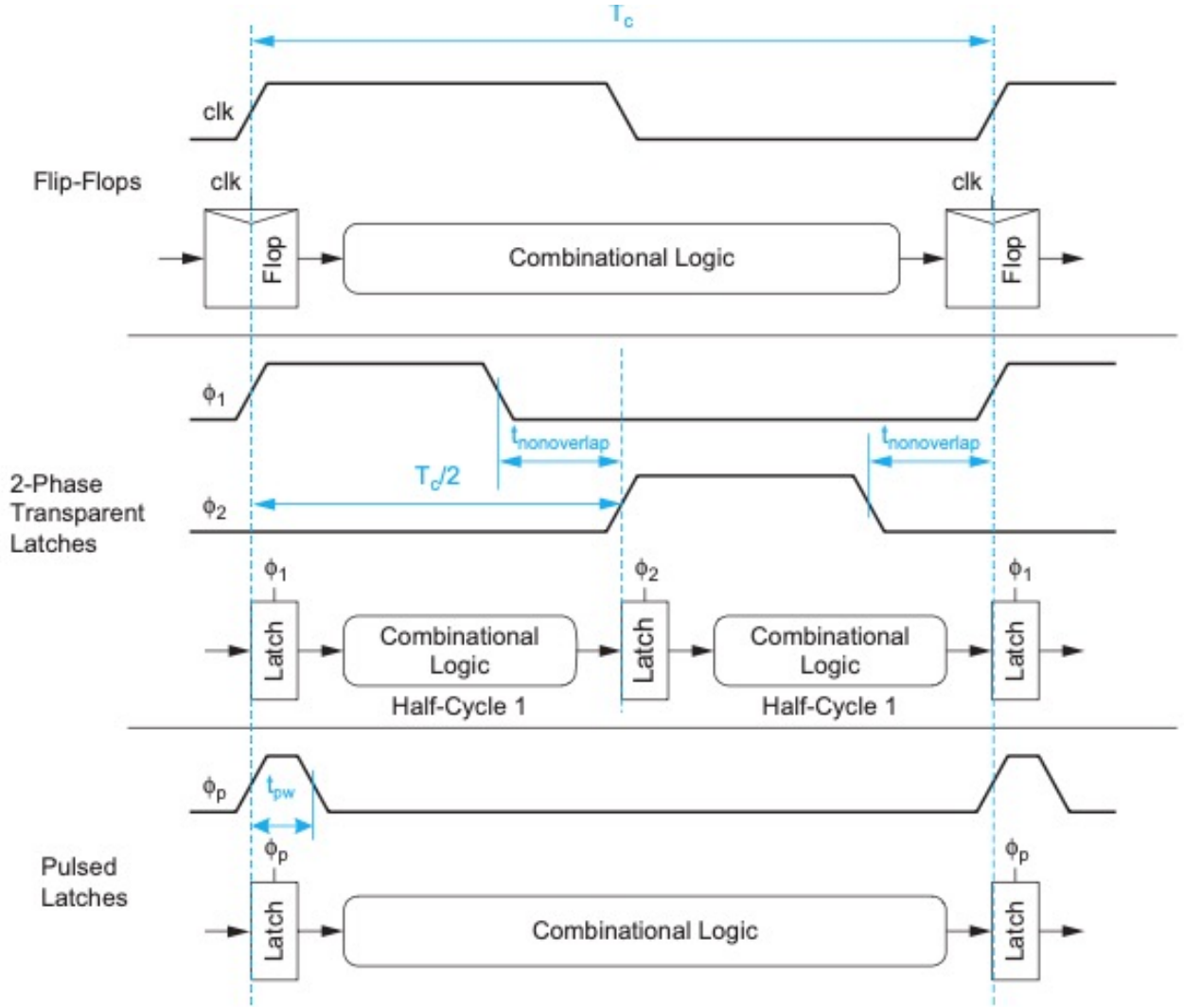
URL: https://elec50010-iac.github.io/course-website/
email: p.cheung@imperial.ac.uk

Latches vs Flip-flops



W&H p376

Clocking Styles

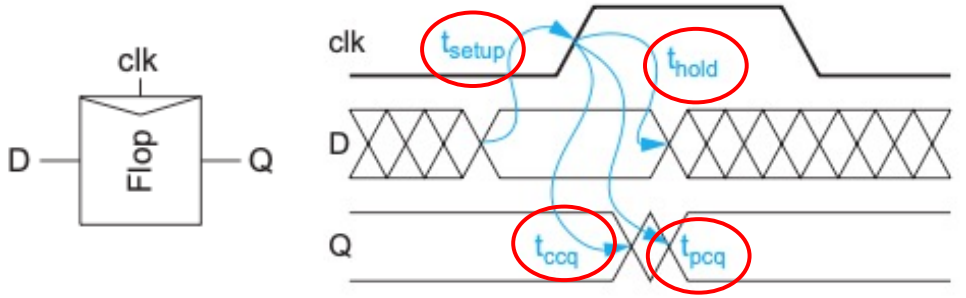


W&H p377-378

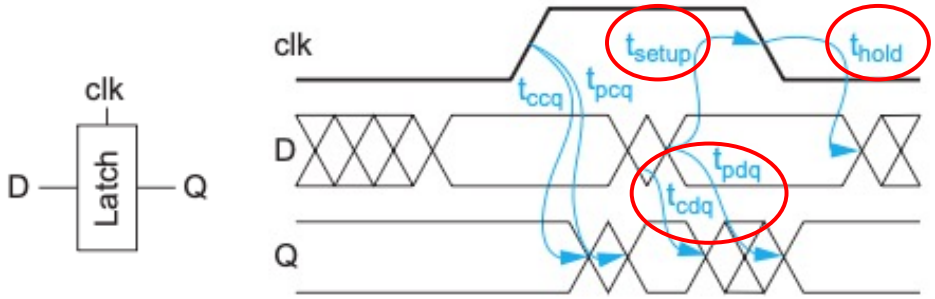
Timing considerations for sequential circuits



(a)



(b)

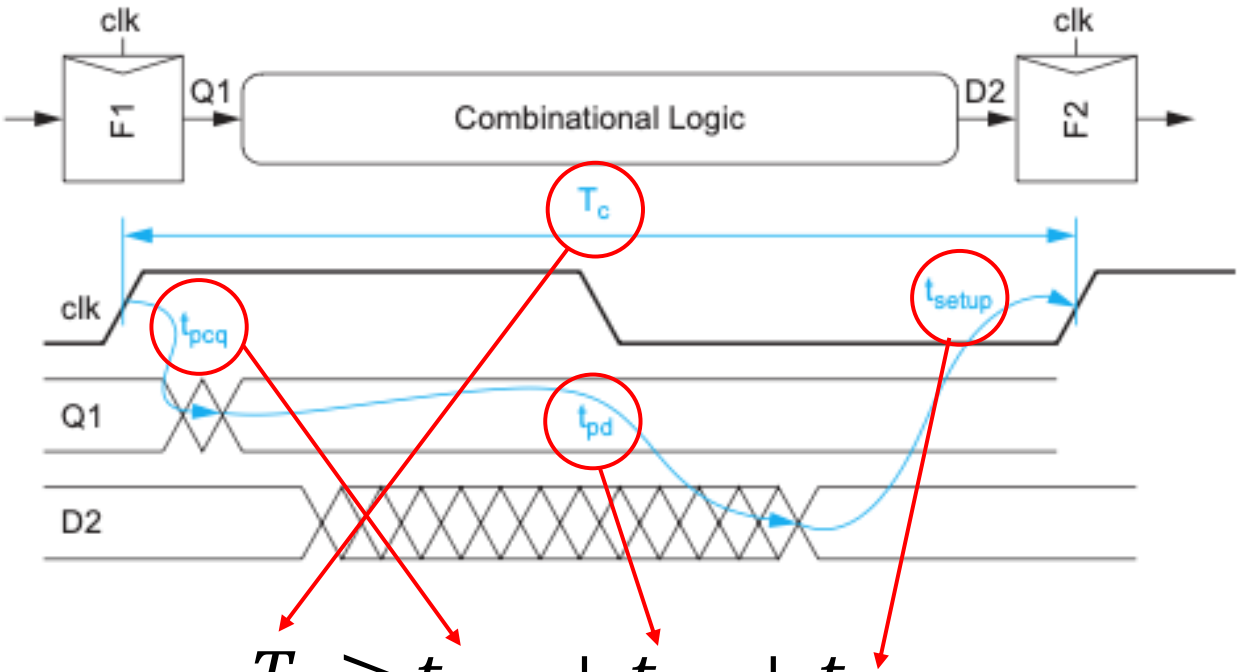


(c)

Term	Name
t_{pd}	Logic Propagation Delay
t_{cd}	Logic Contamination Delay
t_{pcq}	Latch/Flop Clock-to-Q Propagation Delay
t_{ccq}	Latch/Flop Clock-to-Q Contamination Delay
t_{pdq}	Latch D-to-Q Propagation Delay
t_{cdq}	Latch D-to-Q Contamination Delay
t_{setup}	Latch/Flop Setup Time
t_{hold}	Latch/Flop Hold Time

W&H p378-379

Flip-flop Maximum Delay Constraint

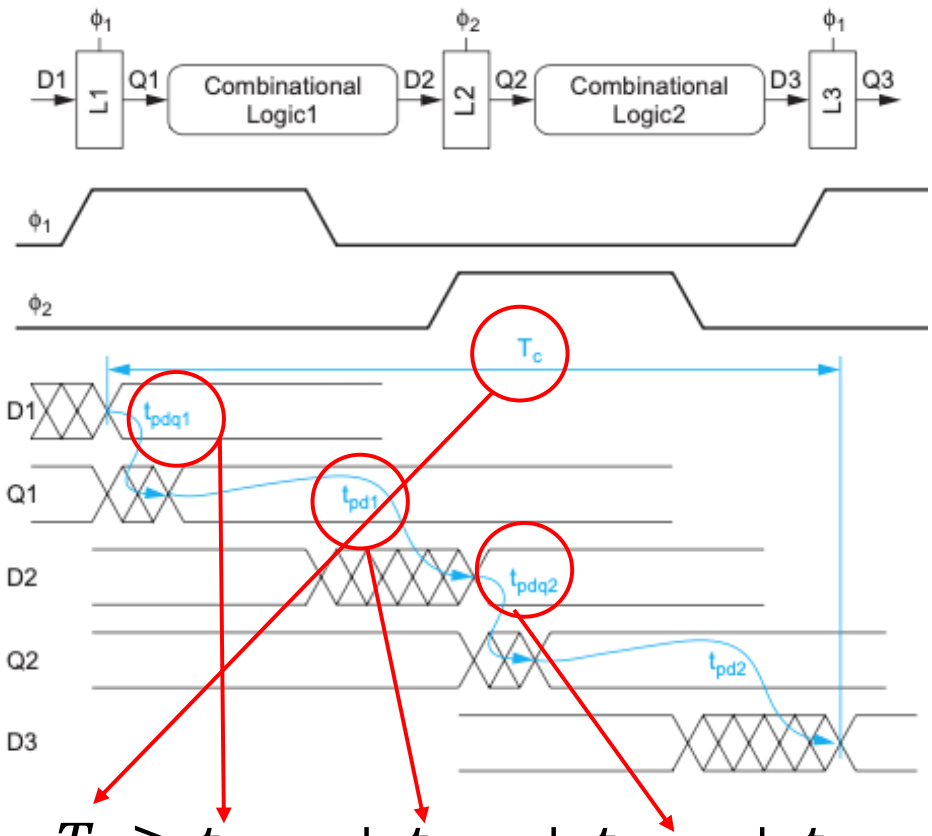


$$T_c \geq t_{pcq} + t_{pd} + t_{setup}$$

$$t_{pd} \leq T_c - \underbrace{(t_{setup} + t_{pcq})}_{\text{sequencing overhead}}$$

W&H p380

Latch Maximum Delay Constraint



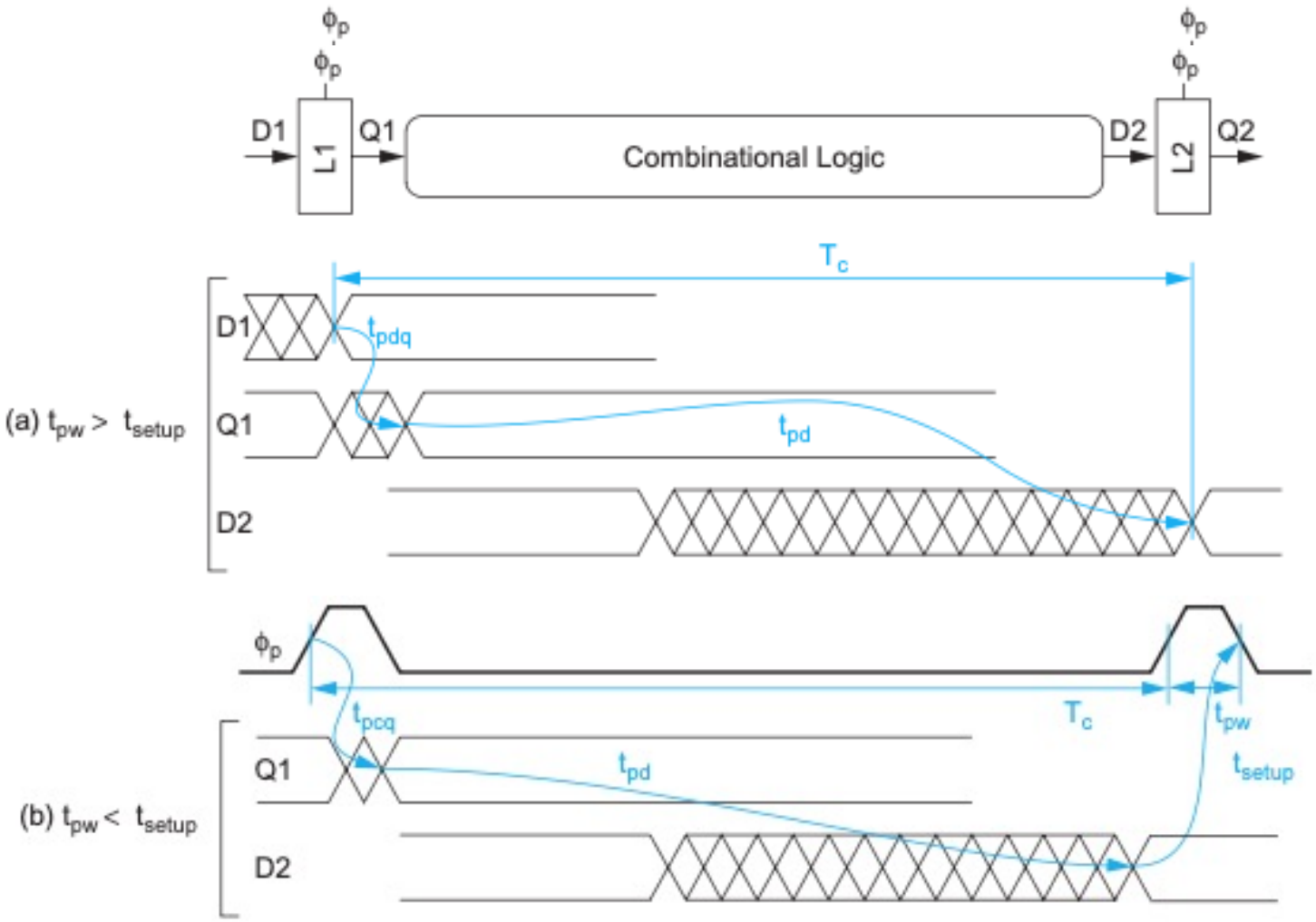
$$T_c \geq t_{pcq1} + t_{pd1} + t_{pcq2} + t_{pd2}$$

$$t_{pd} \leq T_c - (2t_{pcq})$$

sequencing overhead

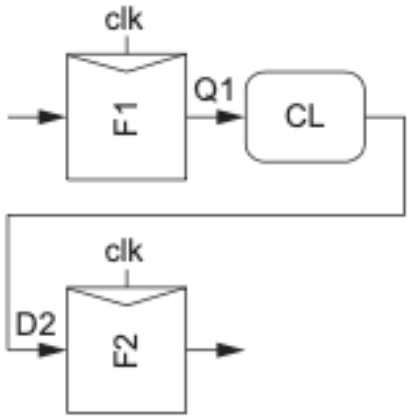
W&H p382

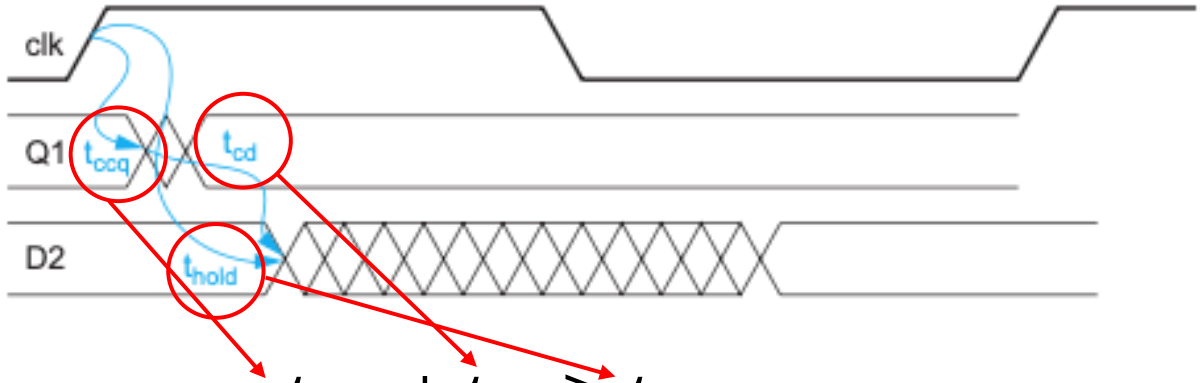
Pulsed Latch Maximum Delay Constraint



W&H p382-383

Flip-flop Minimum Delay Constraint



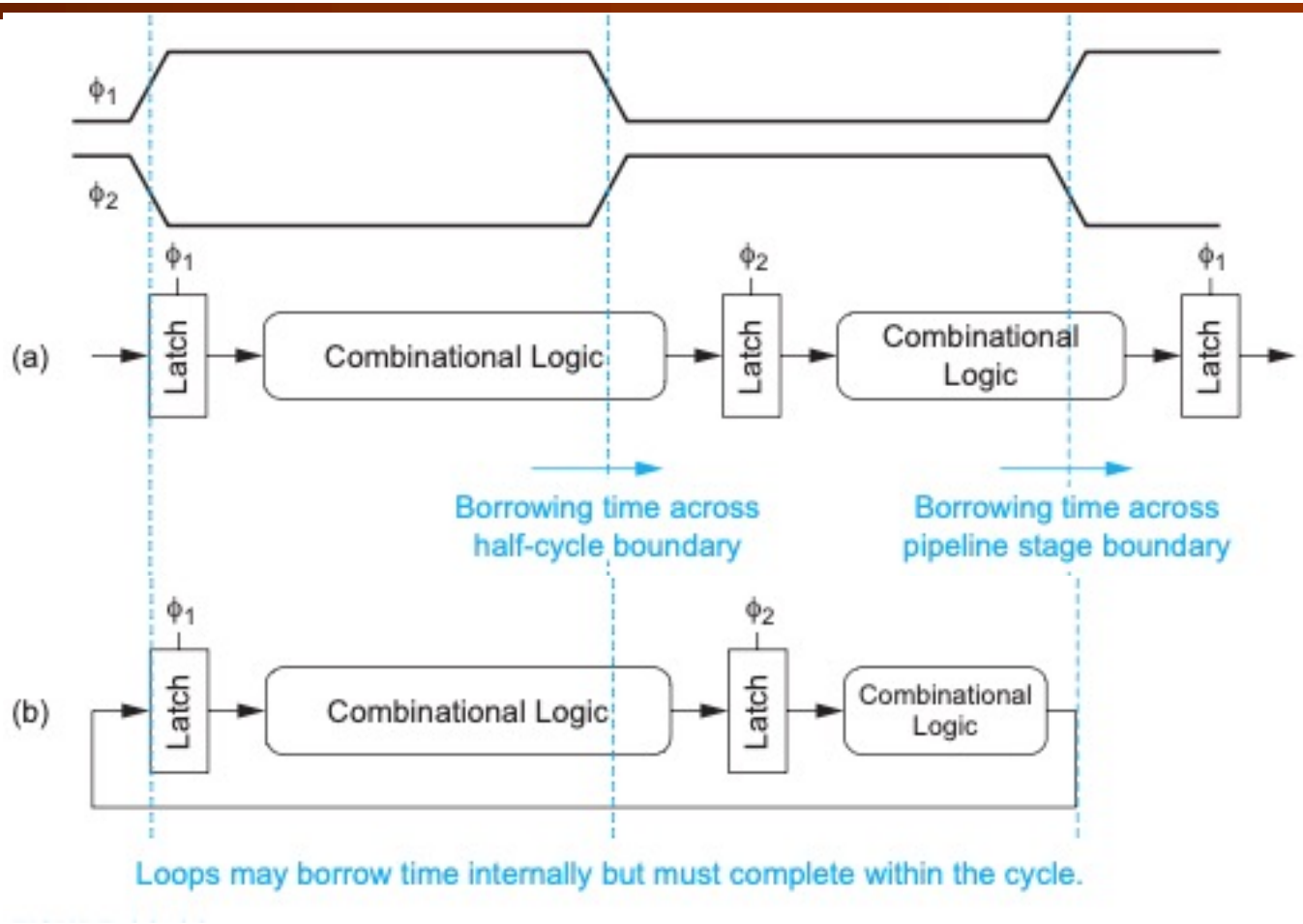


$$t_{ccq} + t_{cd} \geq t_{hold}$$

$$t_{cd} \geq t_{hold} - t_{ccq}$$

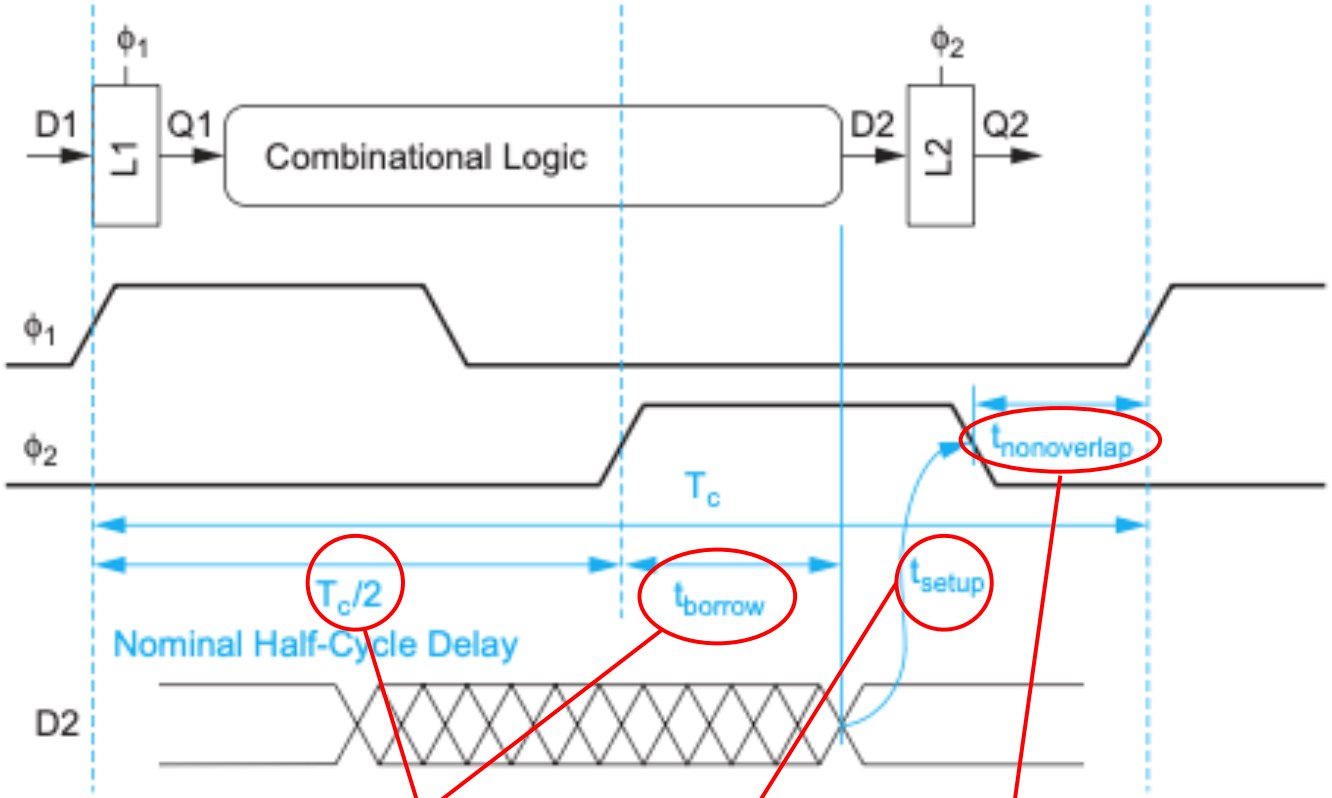
W&H p384-386

Time Borrowing (1)



W&H p387

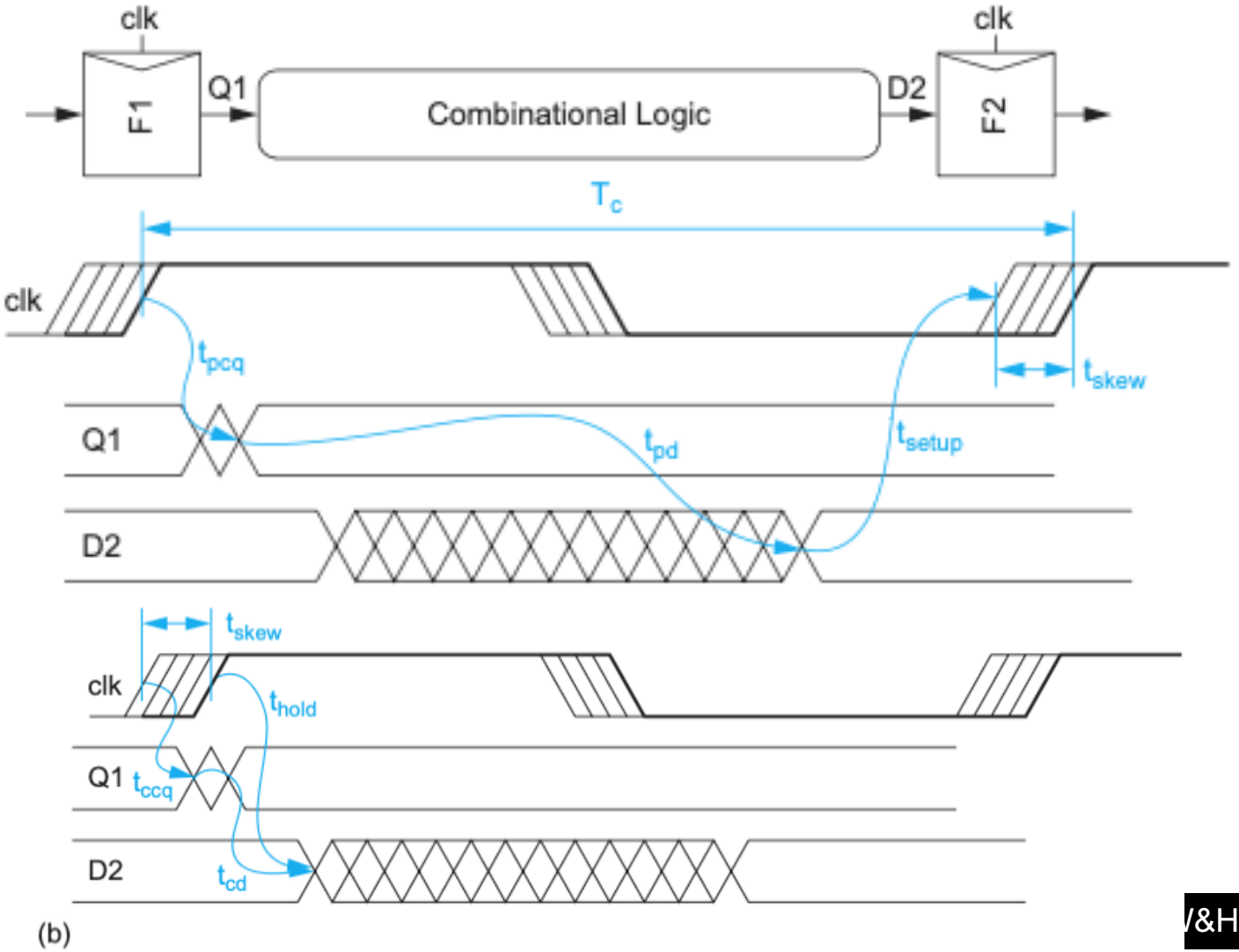
Time Borrowing (2)



$$t_{borrow} \leq \frac{T_c}{2} - (t_{setup} + t_{nonoverlap})$$

W&H 387-389

Clock Skew Problem in Flip-flop

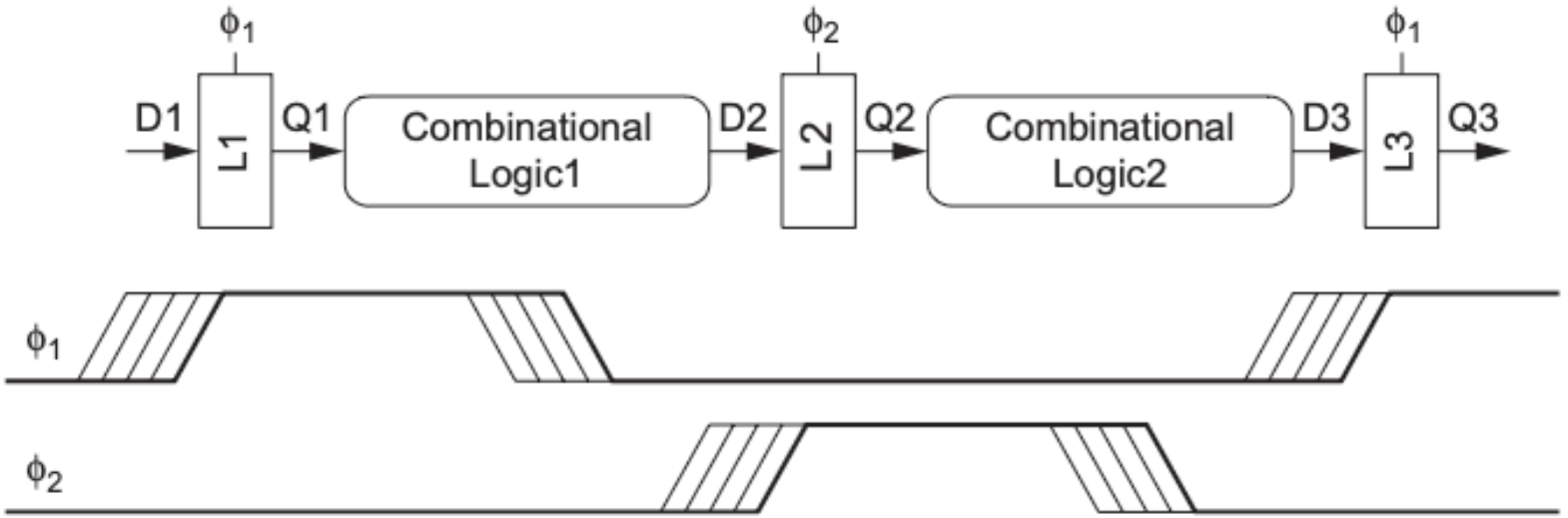


max delay

min delay

V&H 387-389

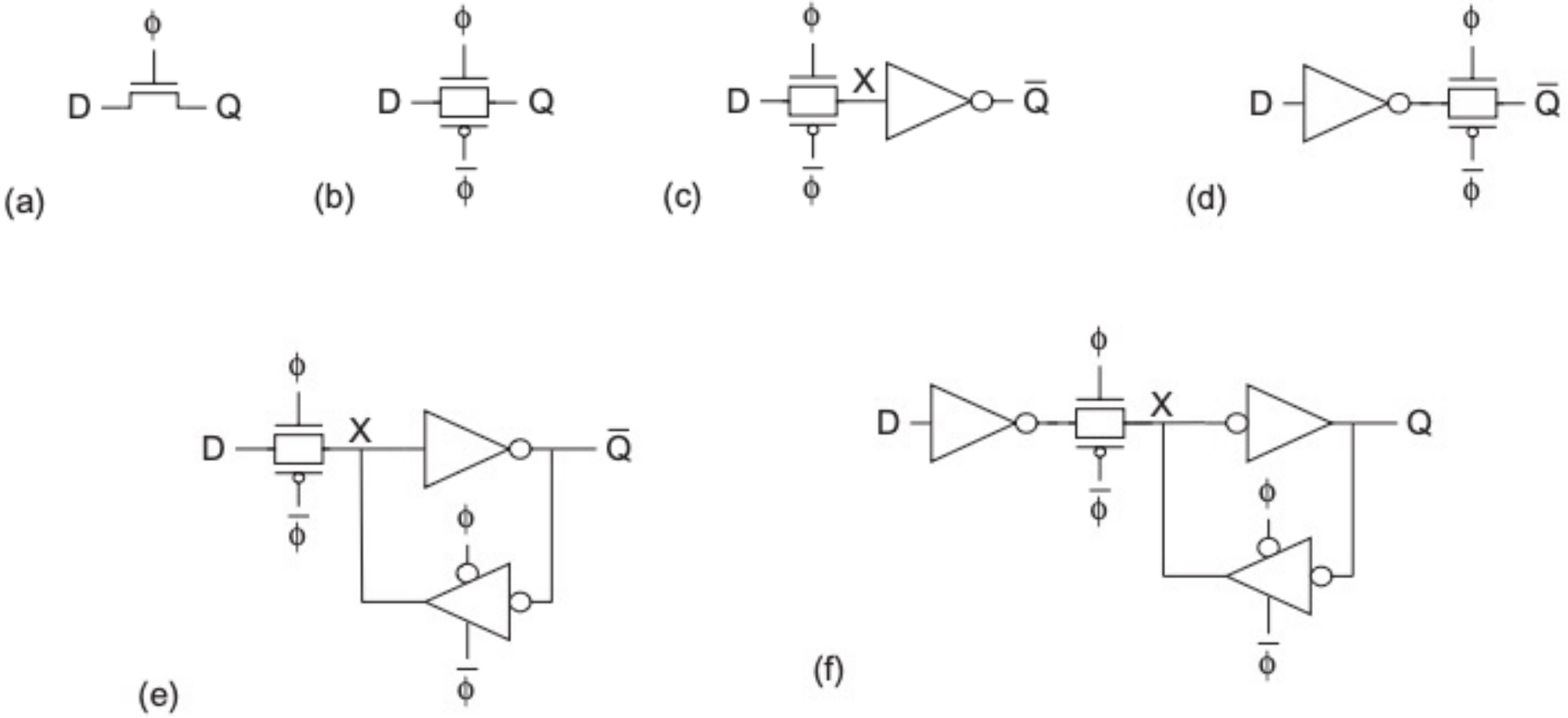
Clock Skew Problem in Transparent Latch



- Transparent latches are **skew tolerant**.

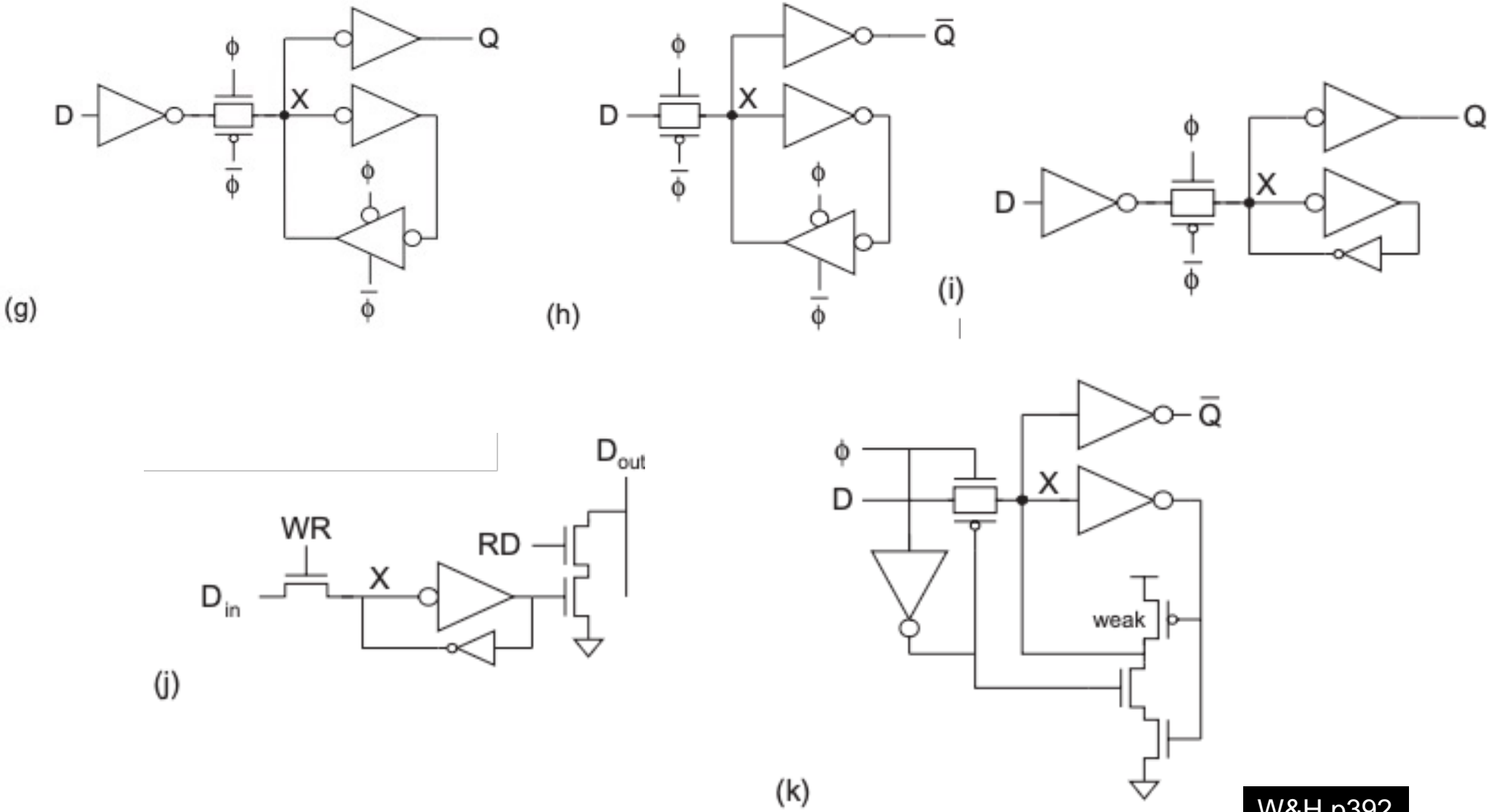
W&H 387-389

Transparent Latch Circuits (1)



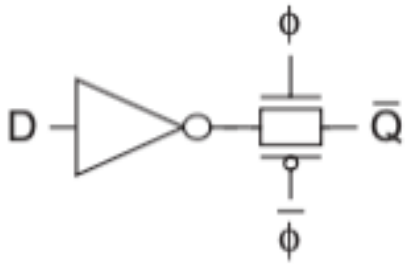
W&H p392

Transparent Latch Circuits (2)

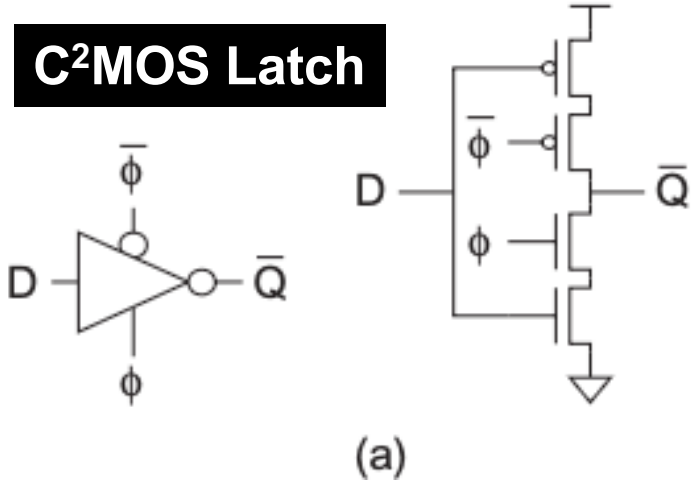


W&H p392

C²MOS Latch Circuit

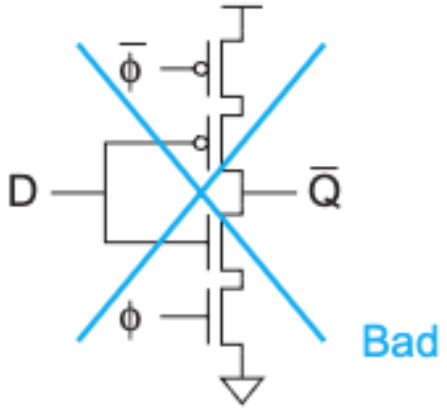


C²MOS Latch



(a)

❑ Charge sharing problem.

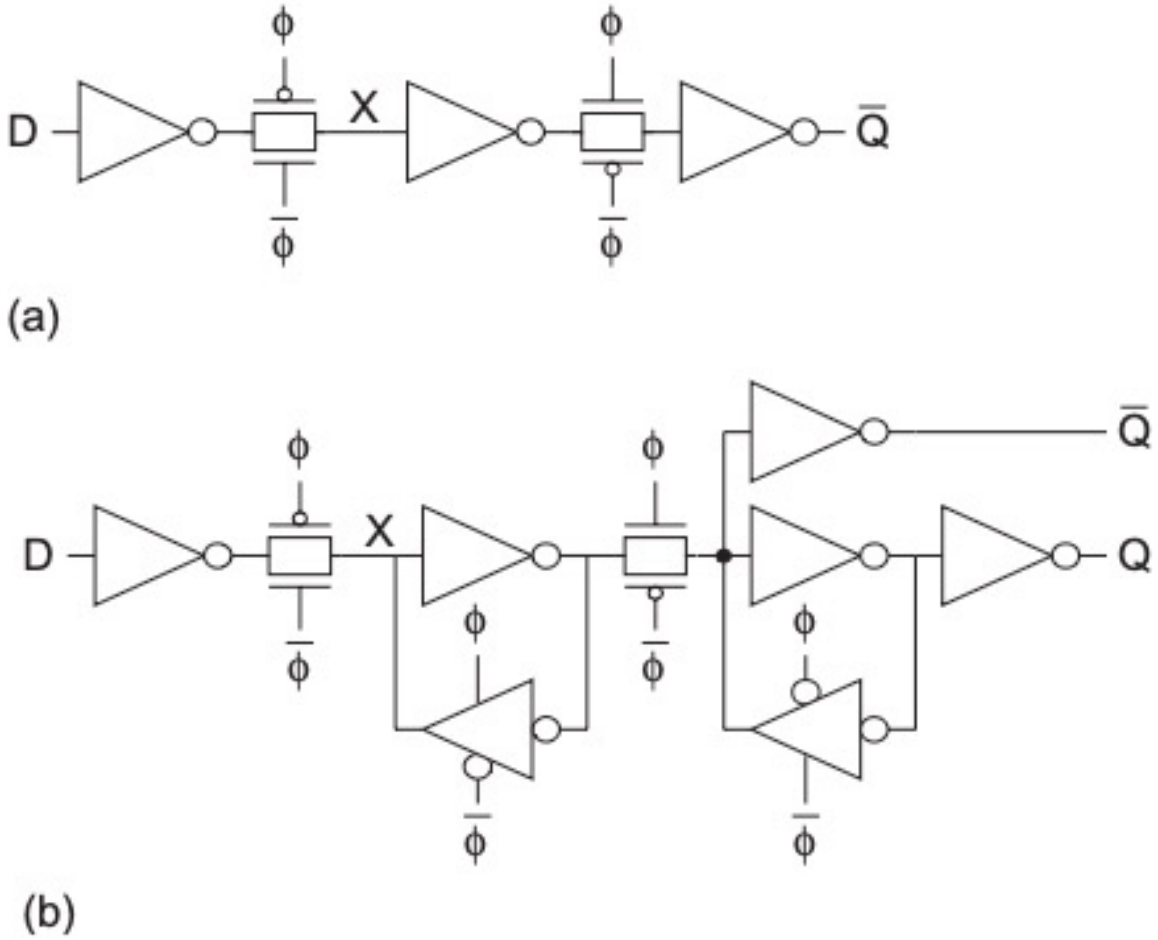


(b)

❑ Clocked CMOS Latch = a tri-state inverter

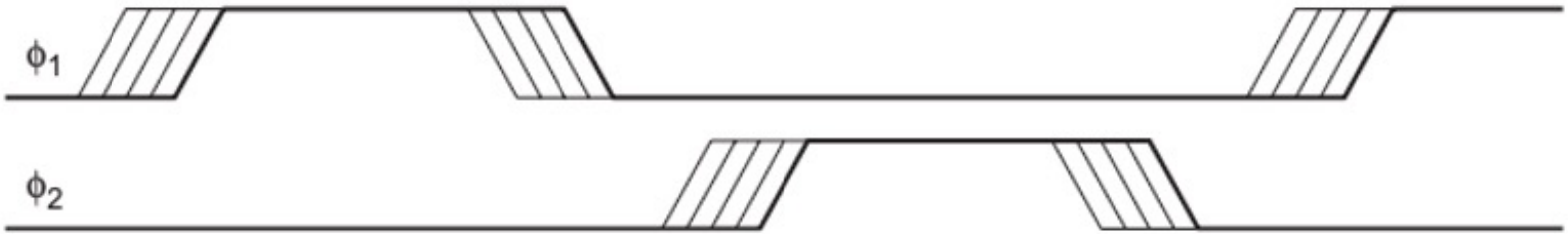
W&H p393

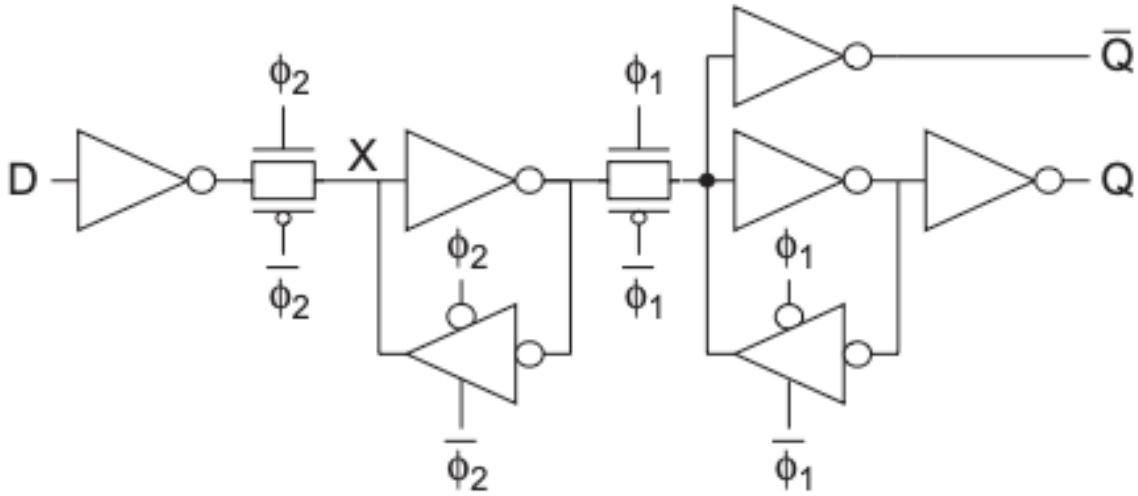
Simple Flip-flop – single phase clock



W&H p393

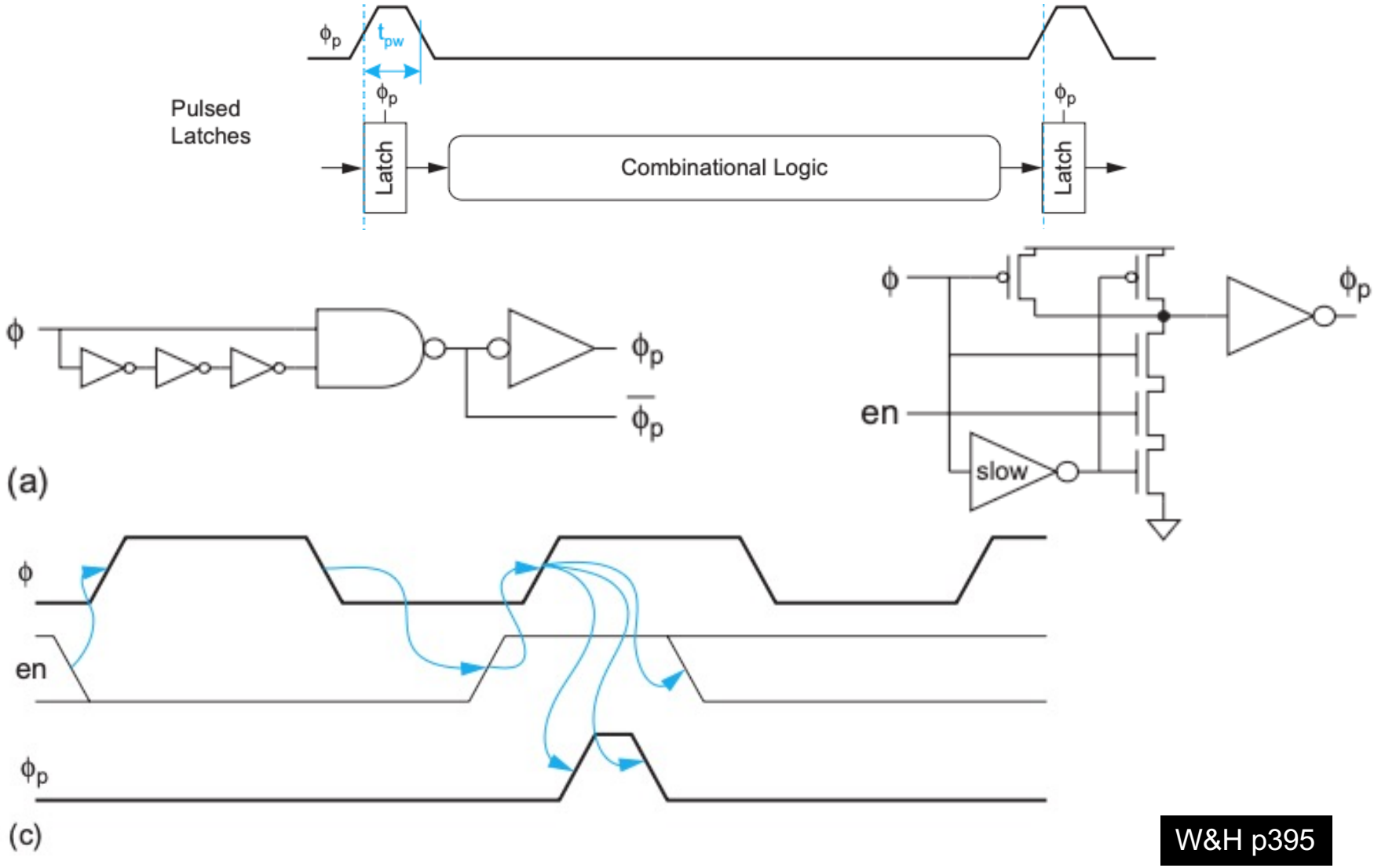
Two-phase clocking Flip-flop





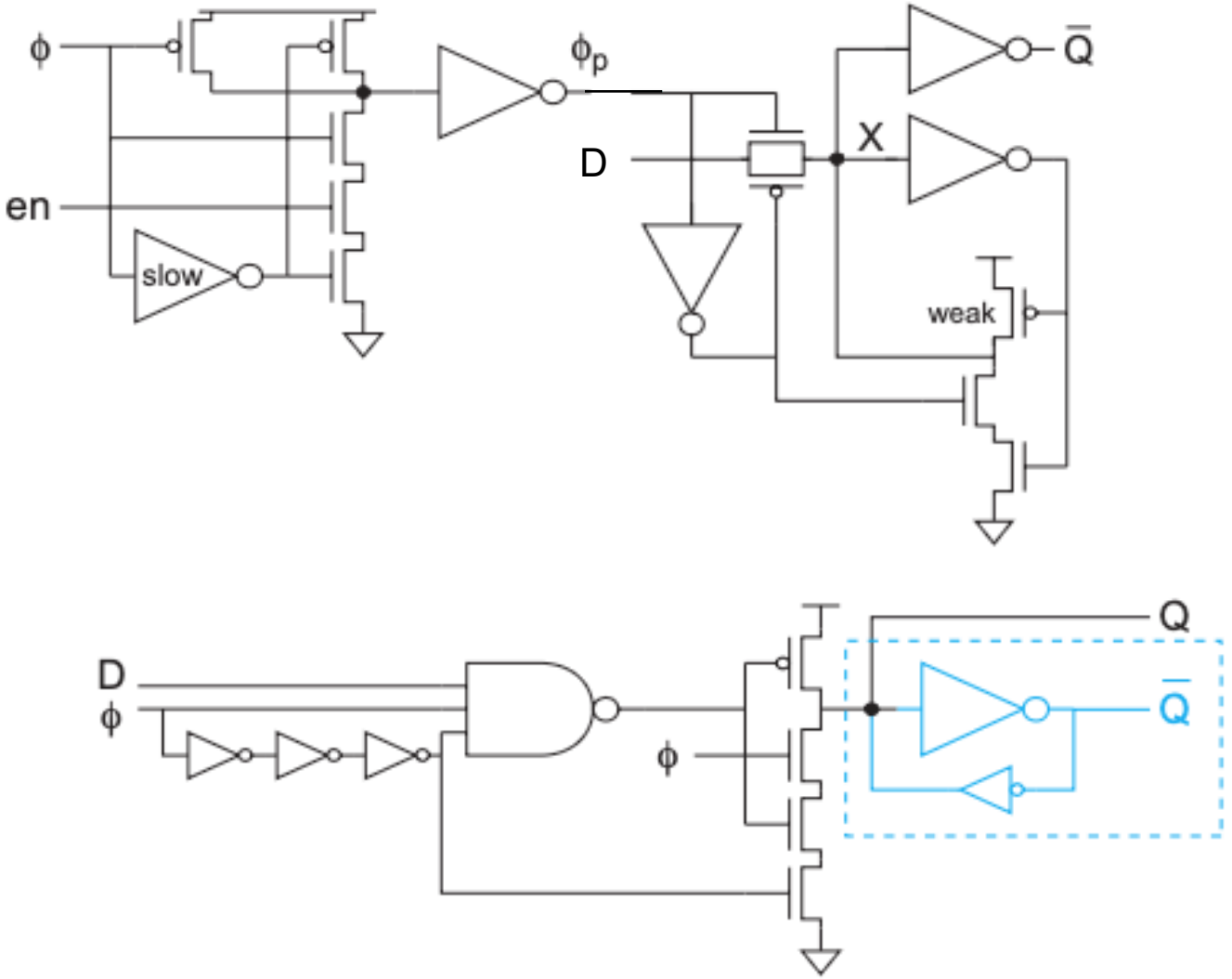
W&H p394

Pulse Generators



W&H p395

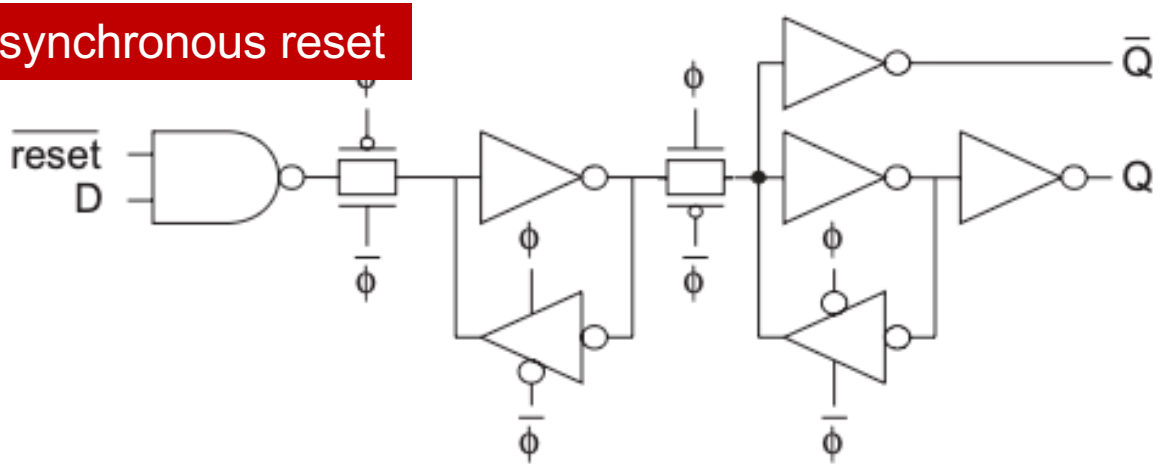
Pulse Latch Circuits



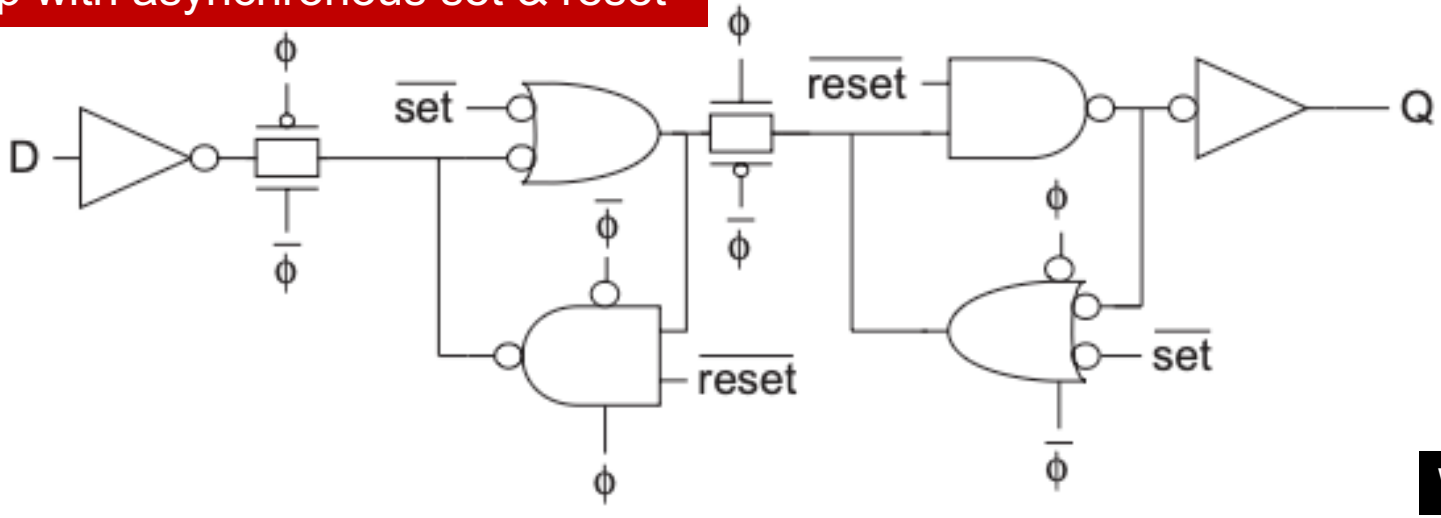
W&H p396

Reset Flip-flop & Latches

Flip-flop with synchronous reset

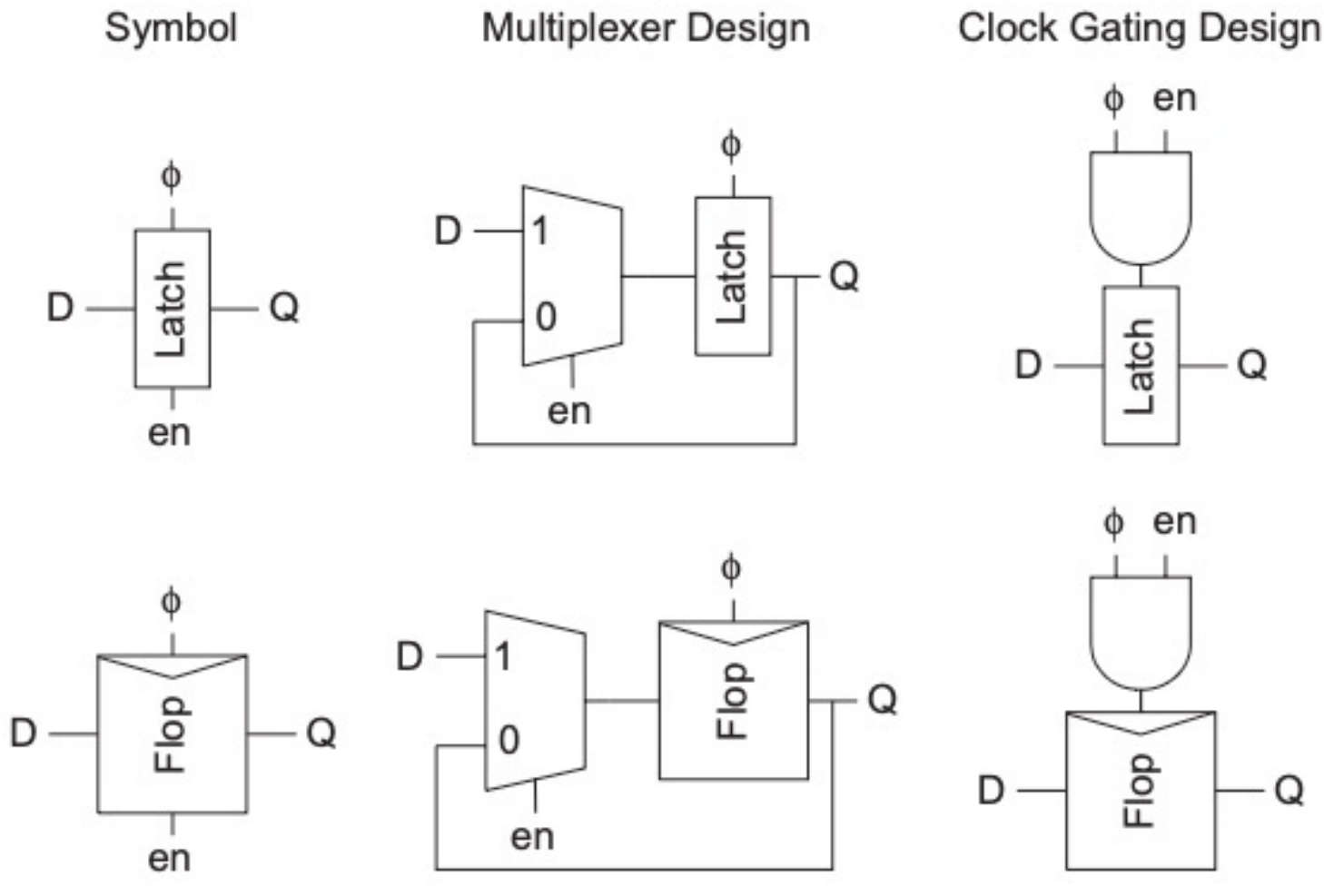


Flip-flop with asynchronous set & reset



W&H p397

Flip-flops & Latches with enable



W&H p398

Advantage of Latches Over Flip-Flops

- If you are going to use clk and clk_b and control skew, why not go back to flip-flops?
- Many people do:
  - Most designs in industry are based on flip-flops
  - Very easy to verify timing
  - Each path between flip-flops must be less than cycle time
  - Tools check for skew and hold time violations
  - Short paths are padded (buffers are added to slow down the signals)
  - Skew in flip-flop based systems affects the critical path
- Latch designs are more flexible than a flip-flop design
  - Gives the designer more rope
  - Need to CAD tools to make sure it works
  - Can borrow time to allow a path to be longer than clock period
  - Can tolerate clock skew -- skew does not directly add to cycle time

Disadvantages of two-phase clocking

- Need four clocks in general
  - Need true and complement of both clocks
- Still need low skew for good performance
  - The skew increases the cycle time of the machine
  - Need low skew between all the clocks for good performance
  - Want to have Φ1 and Φ2 close to coincident
- Many systems use clock and its complement instead of 2 phases
  - Needless to say they are very careful about clock skew
  - For these systems it is still useful to maintain 2 phase timing types, since it ensures you connect all logic to the right latches
  - Call Clk - Φ1 and Clk - Φ2, and go from there.